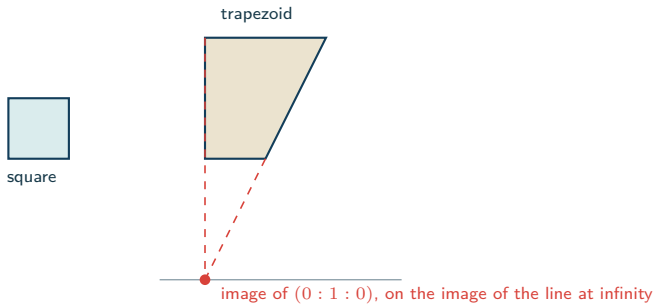
PROJECTIVE GEOMETRY LADDER 5 OF 9

Projective maps, matrices and cross-ratio

An invertible 3×3 matrix acts on the triples, moves the line at infinity, and keeps lines and cross-ratio.

1 / THE RULE AND THE PICTURE

Multiply the triple by a matrix, and parallel lines can converge



A projective map sends $(X:Y:Z)$ to the triple $M(X,Y,Z)^T$ for an invertible matrix M, and rescaling M changes nothing:

$$\boxed{(X:Y:Z) \mapsto M\,(X:Y:Z)^T}$$

With M below, the vertical sides of a square leave parallel and arrive pointing at $(0,-2)$.

2 / THE EXAMPLE

The matrix $M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -\frac{1}{2} & 1 \end{pmatrix}$

A point (x,y) goes to $(x:y:1-\frac{y}{2})$, which is $\left(\frac{x}{1-y/2}, \frac{y}{1-y/2}\right)$, so the corners of the unit square $(0,0),(1,0),(1,1),(0,1)$ land on $(0,0),(1,0),(2,2),(0,2)$. The point at infinity $(0:1:0)$ of the vertical direction goes to $(0:1:-\frac{1}{2}) = (0,-2)$, where the vertical sides now meet, and the whole line at infinity goes to the horizontal line $y=-2$. Horizontal lines stay horizontal, since the point $(1:0:0)$ is fixed.

3 / WHAT IS KEPT AND WHAT IS LOST

Projective geometry studies what survives such maps

property	under a projective map	example
straight lines	kept	a line maps to a line
incidence	kept	points on a line stay on a line
cross-ratio	kept	$(0,1;2,3) = \frac{4}{3}$ becomes $(1,\frac{1}{2};\frac{1}{3},\frac{1}{4}) = \frac{4}{3}$
conics	kept, as a conic	a circle may become an ellipse or a hyperbola
parallelism	changed	parallel lines may meet in a finite point
distance and angle	changed	nothing measured is kept

4 / FOUR POINTS FIX THE MAP

Eight numbers are needed and four points supply them

A 3×3 matrix up to scale has $9-1=8$ free numbers, and each point requires 2 conditions, so four points, no three collinear, can be sent to any other four by exactly one projective map.
This is why a photograph of a rectangle, whose four corners are known, can be straightened by a single matrix.

5 / THE LINE, AND CROSS-RATIO

On the projective line the maps are $x \mapsto \frac{ax+b}{cx+d}$

The matrix $\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ sends x to $\frac{1}{x+1}$ and takes $0,1,2,3$ to $1,\frac{1}{2},\frac{1}{3},\frac{1}{4}$. The cross-ratio $(a,b;c,d) = \frac{(c-a)(d-b)}{(c-b)(d-a)}$ equals $\frac{2\cdot 2}{1\cdot 3} = \frac{4}{3}$ before and $\frac{(-\frac{2}{3})(-\frac{1}{4})}{(-\frac{1}{6})(-\frac{3}{4})} = \frac{4}{3}$ after, and this single number is what four collinear points keep. Composing two such maps multiplies their matrices, so the maps form a group that does not commute.

6 / TRY THESE

Apply the matrix

(a) Where does M send $(1,0)$? To $(1:0:1)$, so $(1,0)$ is fixed.
(b) Where does M send $(0,-2)$? To $(0:-2:2) = (0:-1:1)$, so $(0,-1)$.
(c) The map $x \mapsto x+1$ takes $0,1,2,3$ to $1,2,3,4$, and the cross-ratio is $\frac{(3-1)(4-2)}{(3-2)(4-1)} = \frac{4}{3}$ again.

ASIDE / PERSPECTIVE

A camera is a projective map from space to the plane, which is why parallel railway lines in a photograph meet at a vanishing point.

LINKS TO OTHER SHEETS

The algebra ladder shows why matrix products do not commute, which applies to these maps, and the linear algebra ladder develops matrices as transformations.

Convention. M is an invertible 3×3 matrix; cross-ratio of four collinear points is $(a,b;c,d)$; a projective map is determined by images of four points no three collinear.

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